

WASP-35b

R-1396

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-35_211220 · created 2026-08-01T03:34:52+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2459568.8011 +/- 0.0023 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1646 +/- 0.0073
Transit depth (Rp/Rs)^2	2.71 +/- 0.24 [%]
Orbital Inclination (inc)	87.79 +/- 0.98
Airmass coefficient 1 (a1)	1.068 +/- 0.013
Airmass coefficient 2 (a2)	-0.0422 +/- 0.0089
Scatter in the residuals of the lightcurve fit is	1.2 %
Best Comparison Star	#3 - [472, 123]
Optimal Method	PSF photometry
Transit Duration (day)	0.1344 +/- 0.005

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.1646 ± 0.0073 vs 0.1241 (deviation 4.64σ)
Duration fit vs published	0.1344 d vs 0.1317 d (deviation 0.46σ)
Mid-time O-C	-2.6 min
Depth SNR	18.9
Residual scatter	1.2 %

Light curve

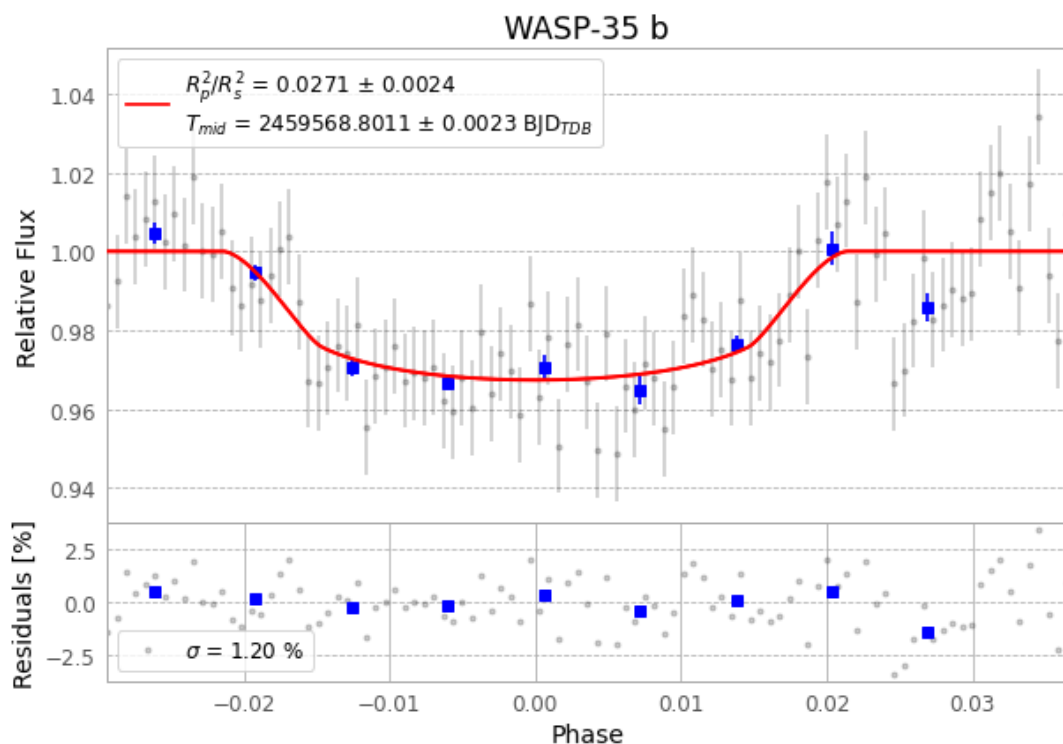


Figure 1 — FinalLightCurve_WASP-35 b_2021-12-19.png (SHA-256 f376f40be6ea36e9...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [388, 155], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 93f653d5d87aa0b4...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

101 FITS frames (66 MB), WASP-35211220045112.FITS → WASP-35211220095112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-35-211220.log (0c2f43a68f32...), FinalLightCurve_WASP-35 b_2021-12-19.png (f376f40be6ea...), FinalLightCurve_WASP-35 b_2021-12-19.csv (0f78074f0a05...)

Compute resources

Wall time 160.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 03:34Z.